Vishnu Vardhan Addanki Tirumala

Curriculum Vitae · MSc Artificial Intelligence & Data Science · University Research Assistant

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RESEARCH INTERESTS

Interpretability and robustness of vision-language models; multimodal learning and parameter-efficient fine-tuning (LoRA); interpretable and distributed learning (Tsetlin Machines); lightweight, sensor-agnostic perception for edge and robotic platforms; agentic AI systems.

EDUCATION

AUG 2025 — PRESENT
MTech in Artificial Intelligence & Data Science
IIITDM & University of Agder · Grimstad, Norway
Research track: vision-language models, interpretable ML and edge perception.

JUL 2021 — JUN 2024
B.E. in Computer Science (AI & ML)
Neil Gogte Institute of Technology · Hyderabad, India

JUL 2017 — MAY 2021
Diploma in Embedded Systems
Government Institute of Electronics · Hyderabad, India

APR 2017
Secondary School Certificate (SSC)
St. Anthony's High School · Hyderabad, India

PUBLICATIONS

2026
Clause Sanitization for Compact and Interpretable Tsetlin Machines
International Symposium on the Tsetlin Machine (ISTM) (Accepted)

MANUSCRIPTS UNDER REVIEW & IN PREPARATION

Communicated
Where does LoRA fall short in multimodal learning?
CVPR 2027

Communicated
Interpretable machine learning for distributed systems
IEEE Transactions on Artificial Intelligence

Communicated
A lightweight, sensing-agnostic segmentation network for the edge
Int. Journal of Applied Earth Observation and Geoinformation

In preparation
Real-time terrain classification for humanoid robots
IEEE Internet of Things Journal

RESEARCH PROJECTS

Communicated
CVPR 2027

Where does LoRA fall short in multimodal learning?
Master's thesis · CVPR 2027
Studied how LoRA fine-tuning affects video-language models (Qwen, SmolVLM), ran controlled diagnostics to find where it falls short, and implemented methods to test whether the problem can be fixed.

Communicated
IEEE TAI

Interpretable machine learning for distributed systems
Research project · IEEE Transactions on Artificial Intelligence
Proves when distributed training preserves the accuracy and explainability of a centralized model, and introduces new ways to combine logically distinct learners into one trustworthy decision-maker.

Communicated
JAG

A lightweight, sensing-agnostic segmentation network for the edge
Research project · Int. Journal of Applied Earth Observation and Geoinformation
A small, edge-deployable segmentation network that works across different sensing modalities (RGB, thermal and NIR imagery) without a separate model per sensor.

Planned
IEEE IoT

Real-time terrain classification for humanoid robots
Research project · IEEE Internet of Things Journal
Lightweight deep learning models that classify the terrain or surface under a humanoid robot in real time from onboard sensors, optimized to run on resource-constrained edge hardware.

Ongoing
coursework

Agentic AI in a mechatronics building pipeline
Course research project · University of Agder
Integrating LLM agents into a hardware-software co-design pipeline, using existing design software as the agents' environment with human-in-the-loop review built into the workflow.

EXPERIENCE

JAN 2026 — PRESENT

University Research Assistant, University of Agder, Norway

- Research on interpretability and robustness: vision-language models (LoRA, multimodal diagnostics), Tsetlin Machines, distributed interpretable learning and lightweight edge perception.
- Authored two papers submitted to the International Symposium on the Tsetlin Machine (ISTM) 2026 and co-authored a third; "Clause Sanitization for Compact and Interpretable Tsetlin Machines" was accepted for publication.
- Communicated manuscripts to CVPR 2027, IEEE TAI and the Int. Journal of Applied Earth Observation and Geoinformation; preparing an IEEE IoT submission.

JUN 2024 — JUL 2025

Product Development Engineer, Machine Learning, Phenom People Pvt Ltd, Hyderabad

- Led development, deployment, and SRE responsibilities for AI scheduling microservices.
- Maintained and optimized a multi-label classification pipeline for multi-agent emails using ML automation and human-in-the-loop review.
- Improved automation accuracy to 95%, lifting candidate email automation from 30% to 48% and recruiter/interviewer automation from 0% to 60%.
- Deployed BERT-based classifiers and built an auditing system to capture intents, sub-intents, and entities, backed by performance dashboards.
- Built core logic for a voice-based screening agent handling 1,000+ calls monthly, with a 99.9%-accurate voicemail detection system.
- Developed a questionnaire-builder agent enabling the product team to generate job-specific questionnaires from job descriptions.

JUL 2023 — JUN 2024

Machine Learning Intern, Wadhwani AI, Delhi

- Built clinical NER models to extract medical entities from free-text data for a Clinical Decision Support System.
- Designed UIs for knowledge-base validation, integrating ASR and TTS.
- Co-developed a RAG-based chatbot for 1M+ frontline healthcare workers to answer domain-specific queries.
- Fine-tuned embedding models and linear adapters to reduce hallucinations in low-data settings, and built synthetic data pipelines for evaluation.

MAY 2023 — JUN 2024

Junior Machine Learning Engineer, Omdena AI · Remote

- Selected as one of 90 contributors across two global AI projects.
- Montreal chapter: worked on speech-to-text and text-to-speech modules for an AI-powered dubbing app.
- Toronto chapter: built and tested chatbot flows using the Rasa framework, Flask, and FastAPI.

JUN 2023 — JUL 2023

AI Intern, AICTE · Remote

- Developed a mental fitness tracker to monitor a country's mental-disorder trends using 12 regression and ML models.
- Performed data cleaning, preprocessing, and feature engineering to improve model performance.

NOV 2020 — MAY 2021

Apprentice , Embedded Systems, Electronics Corporation of India Ltd (ECIL), Hyderabad

- Six-month sandwich apprenticeship under the Apprentices Act with ECIL's Radar & Power Devices division, a Govt. of India (Dept. of Atomic Energy) enterprise.
- Managed assigned projects to schedule and contributed to raising both module production output and module accuracy.

SELECTED PROJECTS

- **AI-assisted hardware-software co-design agents** — Coursework (2026): custom LLM agents for a hardware and software co-design pipeline, using existing design software as the agents' environment, with human-in-the-loop review built in.
- **Deep Research Agent** — A LangGraph-based multi-agent research system that plans, decomposes, searches the web/arXiv/Wikipedia in parallel, validates claims, and self-reflects before synthesizing a cited answer — paired with a Vue/Nitro chat UI that streams its reasoning live.
- **AI video generator** — An end-to-end AI filmmaking pipeline that turns a topic or script into a fully-produced short video — AI-written story, consistent characters, narration, generated visuals, TTS voiceover, background music, and a rendered MP4, with a Streamlit UI and a Supabase-backed FastAPI service.
- **Sight-Sphere Communicator** — Team-led assistive device for visually impaired individuals using ESP32-CAM, ultrasonic sensors, and a multimodal LLM on Raspberry Pi for real-time audio navigation instructions.
- **EduBot — AI university assistant** — A RAG-based chatbot for document Q&A and video-lecture interaction, built with LangChain, vLLM, and transformers, with multi-cloud deployment across Azure, AWS, Render, and Vercel.
- **MedIntelBot — A Local RAG-Powered Medical Assistant** — A locally hosted RAG application delivering medical insights from real medical PDFs — no cloud-based LLM APIs. 🔧 Built with LangChain, FAISS, CTransformers, and Chainlit for speed, privacy, and precise document retrieval.
- **Face recognition attendance system** — Bachelor's mini project: a Django attendance system that recognizes students live from webcam feeds, using OpenCV DNN face detection, OpenFace embeddings, and a custom-trained classifier — I led the team building it end-to-end.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java

Machine learning & deep learning: PyTorch, Transformers, TensorFlow, Keras, scikit-learn

NLP & GenAI: LLMs, LangChain, LangGraph, RAG, spaCy, NLTK, Prompt engineering

Frameworks: FastAPI, Flask, Django, Streamlit, Gradio, Rasa, Chainlit, React

Databases & vector stores: PostgreSQL, MongoDB, Redis, Elasticsearch, Weaviate, MySQL

DevOps & infra: Docker, Kubernetes, Jenkins, ArgoCD, Kafka, RabbitMQ, Git, Linux

HONOURS & AWARDS

- **NFS winner — Neil Gogte Institute of Technology** — Placed repeatedly in NGIT's weekly NFS coding contest: 3rd in week 2, 4th in week 7, and 5th in week 4.
- **Flash mob — runner-up & special dance prize** — Runner-up in an inter-college flash mob competition, and awarded a special prize for dance.

SERVICE & ACTIVITIES

- **Co-Head, Dance Club**, Neil Gogte Institute of Technology · Hyderabad, India (OCT 2021 — APR 2024)
- **Cadet, National Cadet Corps (NCC)**, India (AUG 2015 — MAY 2020)

LANGUAGES

Telugu and Hindi (native); English (professional).

